

April'26 Major Examination: EEL71020 Hardware Design for AI (OPEN-BOOK)

Guidelines (Total time: 180 minutes, Maximum Marks: 54):

- Please read the question paper very carefully. **NO clarification is required in any question.** In case of any doubt, assume whatever you wish to and state that in your answer.
 - Usage of LLM/AI tool is NOT allowed. BOOKS/NOTES etc. can be referred to. For Offline students, access to LAPTOP/MOBILE is NOT allowed.
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1. You are designing an ASIC accelerator for a transformer-based inference engine. The architecture consists of three parallel self-attention units followed by a majority voting unit to improve robustness.

- Query, Key, Value projections using matrix multiplication:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

- Attention output:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d}} \right) V$$

Hardware details are as following: Each MAC (Multiply-Accumulate) unit performs 1 operation per cycle. Each self-attention block has 256 MAC units. Softmax and scaling take 200 cycles per sequence. Memory access latency per matrix load = 100 cycles. Majority Voting Unit details are as following: Takes outputs from 3 attention blocks (each produces a vector of size 64×128). For each element, voting takes 2 cycles. Voting is fully parallel across 128 columns, but sequential across 64 rows.

(a) Compute total number of MAC operations required for one self-attention unit (including Q, K, V projections and attention computation QK^T and final multiplication with V). (b) Compute the total latency when three self-attention units operate in parallel. (c) With a relevant diagram, derive the expression and compute the overall system latency for the full operation (attention + voting). [8]

2. You are leading the design of a custom ASIC accelerator for a CNN inference engine that exploits weight sparsity using the Compressed Sparse Row (CSR) format to reduce computation and memory overhead. A convolutional layer with: Input feature map size: 32×32 , No. of input channels: $C_{in} = 16$, No. of output channels: $C_{out} = 32$, Kernel size: 3×3 , Stride = 1, Padding = 1 (output size remains 32×32). It is known that 70% of the weights are zeros. The hardware details are as follows: Accelerator has 128 MAC units, Each MAC processes 1 non-zero multiplication per cycle, Index decoding overhead: 1 extra cycle per non-zero element, and 50 cycles per fetch of CSR data memory access latency per output channel.
- (i) Estimate the total computation cycles required for the convolution layer considering: Parallel MAC units, and Index decoding overhead (ii) Compare the speedup achieved over a dense implementation (assume dense computation uses same 128 MAC units but no sparsity or decoding overhead)? [8]

3. Design a hardware architecture to efficiently implement the following ANN:

Input layer: 1024 neurons, Hidden Layer 1: 512 neurons, Hidden Layer 2: 256 neurons, Output layer: 10 neurons, and Activation function: ReLU for hidden layers, Softmax for output

The accelerator must meet the following constraints:

- i) On-chip SRAM budget: 1 MB ii) External DRAM access is expensive (high latency and power) iii) Target throughput: at least 1000 inferences per second iv) Power budget: < 2 W

Hint: Organize your answer in this manner: propose a suitable data-flow mechanism, design a single MAC unit, propose a parallel arrangement of MACs, design a suitable memory hierarchy structure and an efficient data transfer mechanism to suit this memory hierarchy? [10]

4. You are hired as an intern in a deep-tech startup *PrajnaTech* where you are tasked with designing a re-configurable ASIC/FPGA-based deep learning accelerator capable of efficiently supporting multiple model types used in modern AI workloads: CNN (e.g., image classification), RNN/LSTM (e.g., speech processing), and Transformer (self-attention for NLP). The accelerator must adapt to different computational patterns while maintaining high performance and energy efficiency. The company's upper management has given below system requirements: Peak compute: 1 TOPS, On-chip SRAM: 2 MB, External DRAM bandwidth: limited, Must support dynamic model switching at runtime, and Power budget: < 5 W.
 - (a) Propose a high-level architecture that can support: Convolution operations (CNN), Matrix-vector and sequential operations (RNN/LSTM), and Matrix-matrix and attention operations (Transformers)? [5]
 - (b) Design a control mechanism that: Schedules operations for different model types, Handles dependencies (especially in RNNs and Transformers), and Issues instructions to PEs efficiently? [5]
 - (c) Suggest enhancements to your hardware architecture for improving efficiency across different models: Sparsity support and Operator fusion (e.g., Conv + ReLU, Attention + Softmax)? [4]
5. In May 2025, during India-Pakistan clash, indigenous (locally developed) electronics was felt as a necessity in the modern warfare because of the rampant usage in UAVs, drones etc. Coupled with AI-based techniques, these weapons can turn out to be lethal and hence premier R&D institutions of the country need to develop hardware designs locally to support Indian defense sector. IIT Jodhpur is aggressively working to contribute to country's self-reliance in defense sector. You decide to work in one such project where the goal is to design a high-performance ASIC accelerator for General Matrix-Matrix Multiplication (GEMM) that will serve as the core compute engine for diverse AI workloads (including CNNs, RNNs, and Transformers) in defense-related tasks. System Constraints are provided as follows:
 - Peak compute capability: 4 TOPS
 - On-chip SRAM: 4 MB
 - External DRAM bandwidth: limited to 50 GB/s
 - Supported precisions: FP16 and INT8 (runtime selectable)
 - Matrix sizes vary dynamically: Small: 64×64 , Medium: 512×512 , Large: 4096×4096

What architectural decisions (i.e., parallel execution block design, memory bandwidth optimization etc.) you would undertake to develop the design such that the above constraints are followed? With detailed justification, prove that your design choices fulfill above constraints? [Note: Use Question-3 hints!] [7]

6. Supported by some of your close friends, you decide to incubate a startup at IIT Jodhpur instead of sitting for otherwise uncertain college placements and name your venture as *ThirdEyeTech*. The idea is to put a mini-camera in a smart-watch to enable image capture just by hand gestures and then images are to be analyzed through Deep Learning algorithm. For this purpose, you plan to design an intelligent image processing system (I2PS) with the below system constraints:
 - Target platform: MCU with 256 KB SRAM and 1 MB Flash, No GPU / no external DRAM
 - Camera input: 96×96 grayscale images
 - Power budget: < 200 mW
 - Real-time requirement: ≥ 10 FPS
 - Deployment must be fully on-device (NO cloud inference)
 - (a) Propose a compact CNN-based architecture suitable for I2PS: Number of layers, filter sizes, and feature maps, Whether to use depthwise separable convolutions or standard convolutions? Justify your design choices based on memory and compute constraints as mentioned above.
 - (b) Explain how you will reduce model size to fit within Flash memory? [5+2]